Government College of Engineering, Jalgaon - 425001

[An Autonomous Institute of Government of Maharashtra]

Department of Computer Engineering

A

Project report

On

Virtual Classroom

(AcademicYear2025-26)

Submitted By:

Kirtiman Kadam(PRN:2141029)

Siddhesh Girish Dhok, 2241017

Kasu Gazanfar Akhtar,2241028

Karan Bharat Khandagale, 2241031

Guided by

Prof. S.D.Cheke.

Project-I Review Document (2025–26)

1. Problem Statement & Objectives

Problem Statement:

Traditional classroom activities such as attendance tracking, distribution of study materials, and personalized learning support are time-consuming and inefficient. Teachers face challenges in sharing notes and managing daily activities, while students struggle to organize and revise learning content.

The proposed Virtual Classroom System (website/app) aims to solve these issues by providing:

- A dedicated platform where teachers can upload PDFs, notes, books, and materials.
- A face recognition–based automated attendance system.
- A PDF summarizer allowing students to generate concise summaries for revision.
- A centralized and user-friendly system for both teachers and students.

Objectives:

- To design a web-based virtual classroom portal using Django as the backend.
- To automate attendance using AI-based facial recognition.
- To enable teachers to upload learning materials in various formats.
- To enable students to view materials and generate AI-based PDF summaries.- To create a scalable, secure, and easy-to-use e-learning platform.

2. Literature Survey & Research Gap

Existing Systems:

1. Google Classroom
- Allows material sharing but lacks integrated AI summarization and face-recognition attendance.

2. Microsoft Teams Classroom

- Provides attendance for meetings but is heavy, complex, and not material-centric.

3. Moodle

- Powerful LMS but difficult to set up and lacks AI tools like summarization and automated faceattendance.

Findings from Research:

- AI technologies significantly enhance learning management.
- Face recognition using CNN, OpenCV, and embeddings (FaceNet/dlib) improves attendanceautomation.
- NLP models like T5, BART, and Pegasus assist in generating accurate summaries.

Research Gap:

Current systems do not combine:
- Material sharing
- AI-based summarization
- Face-recognition attendance All in one unified, lightweight platform.

The proposed system addresses this gap using Django backend and integrated AI modules.
3. Proposed Methodology / System Design

System Modules:

A. Teacher Module:
- Upload PDFs, notes, books
- Manage classes and students
- View automated attendance using face recognition

B. Student Module:
- Access uploaded materials
- Generate PDF summaries using AI- Download notes and view updates

C. Admin Module:
- Manage user data
- Monitor system activity- Manage class/course structure

System Architecture (Textual):

Teacher → Upload Materials → Virtual Classroom System (Django Backend)

Student → Access Materials → AI Summary Generator

Face Recognition Module → Attendance Records → Teacher Dashboard

Technology Stack:
Frontend: Streamlit
Backend: Django, DRF
Database: MySQL / Sqlite3
AI for Attendance: OpenCV, lbph
AI Summary: genai, pypdf, gTTs

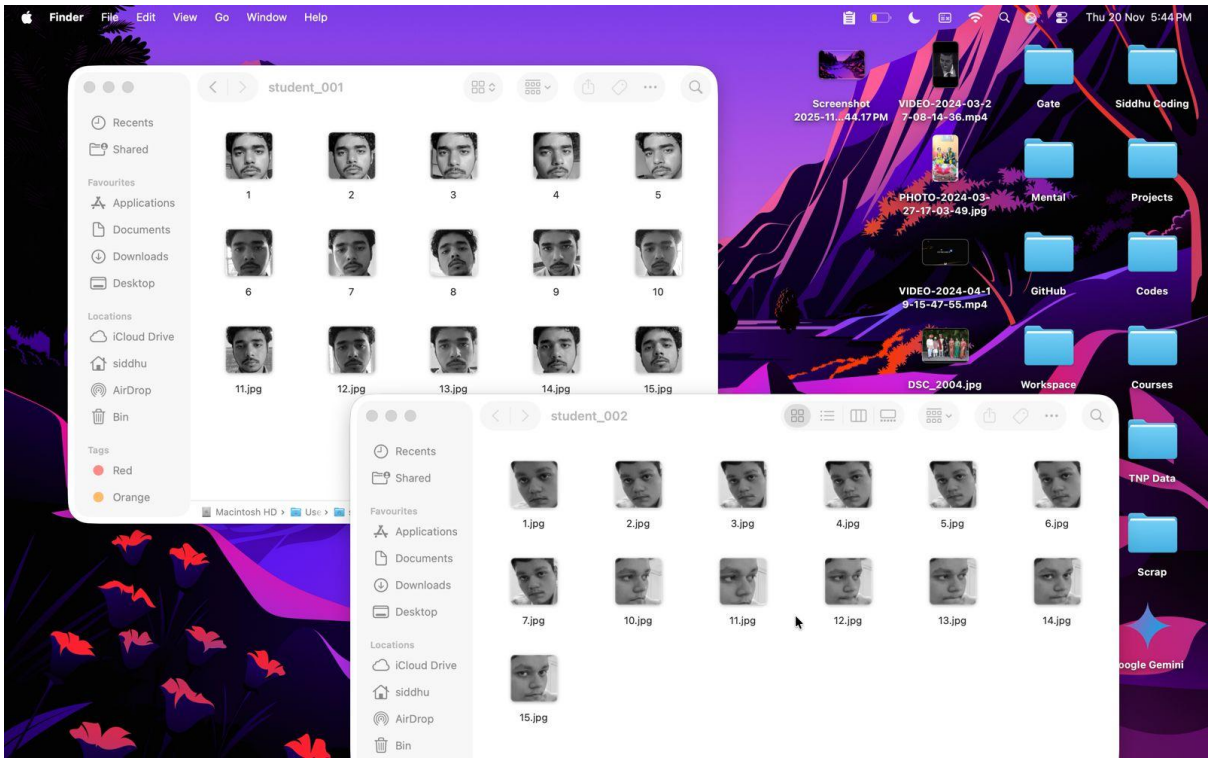
Storage: Cloud or Firebase Storage

5. Future Plan of Work:
- Improve accuracy of face recognition using deep learning methods.
- Fully integrate attendance dashboard within the teacher module.
- Build advanced analytics for attendance and material usage.

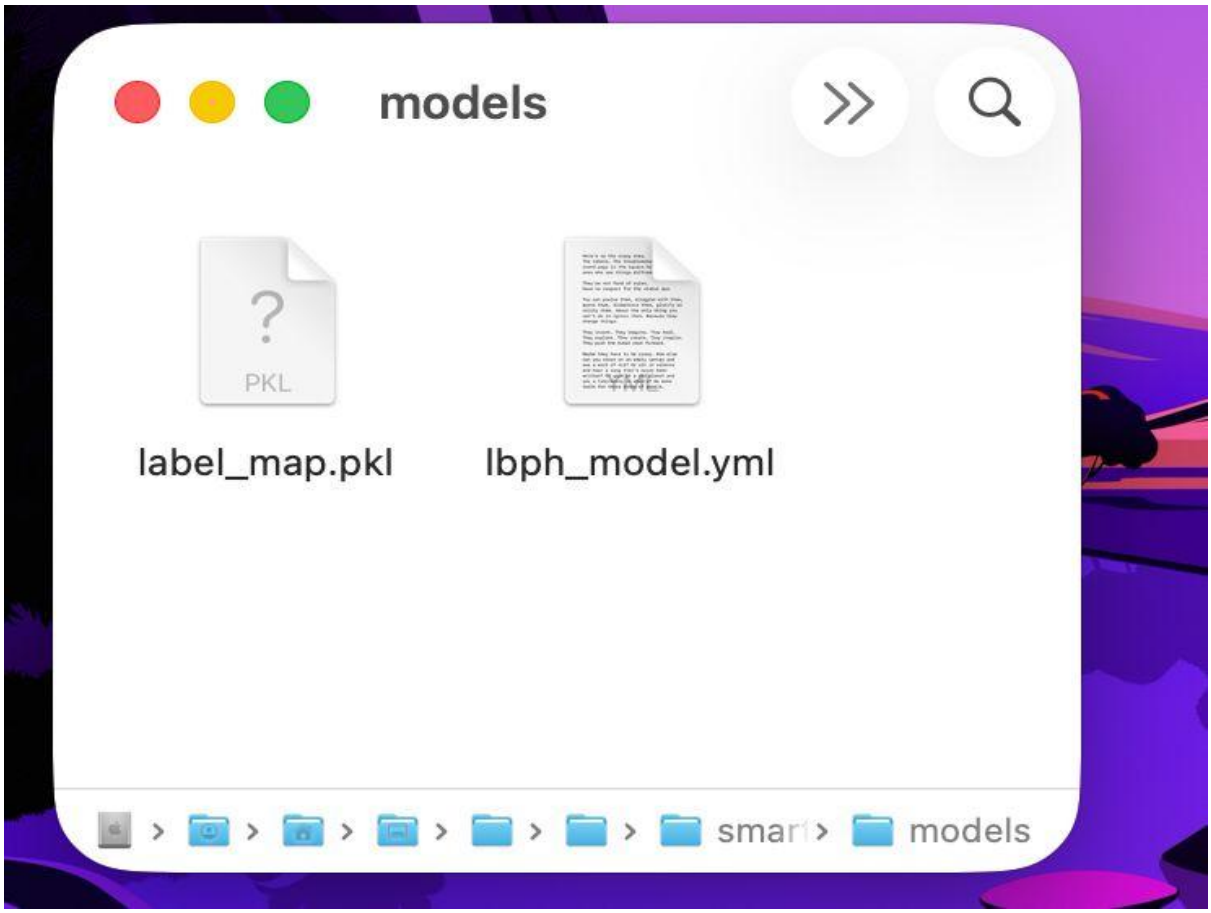
- Enhance PDF summarization using transformer-based NLP models.
- Develop a mobile-friendly user interface for students.
- Implement assignment upload and grading features.
- Add notification systems (email/app notifications).
- Perform end-to-end testing, optimization, and deployment.

1.Attendance using face recognition

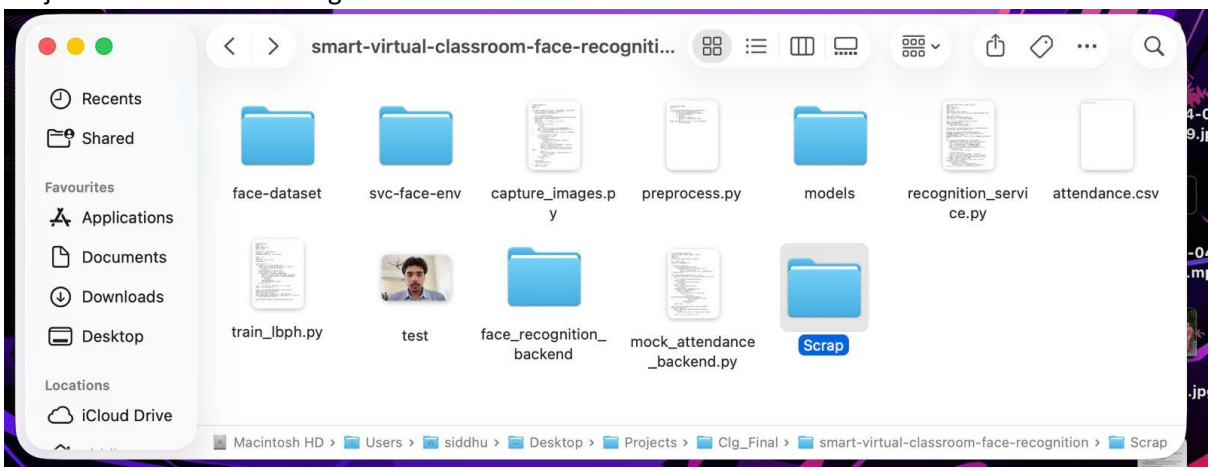
Dataset



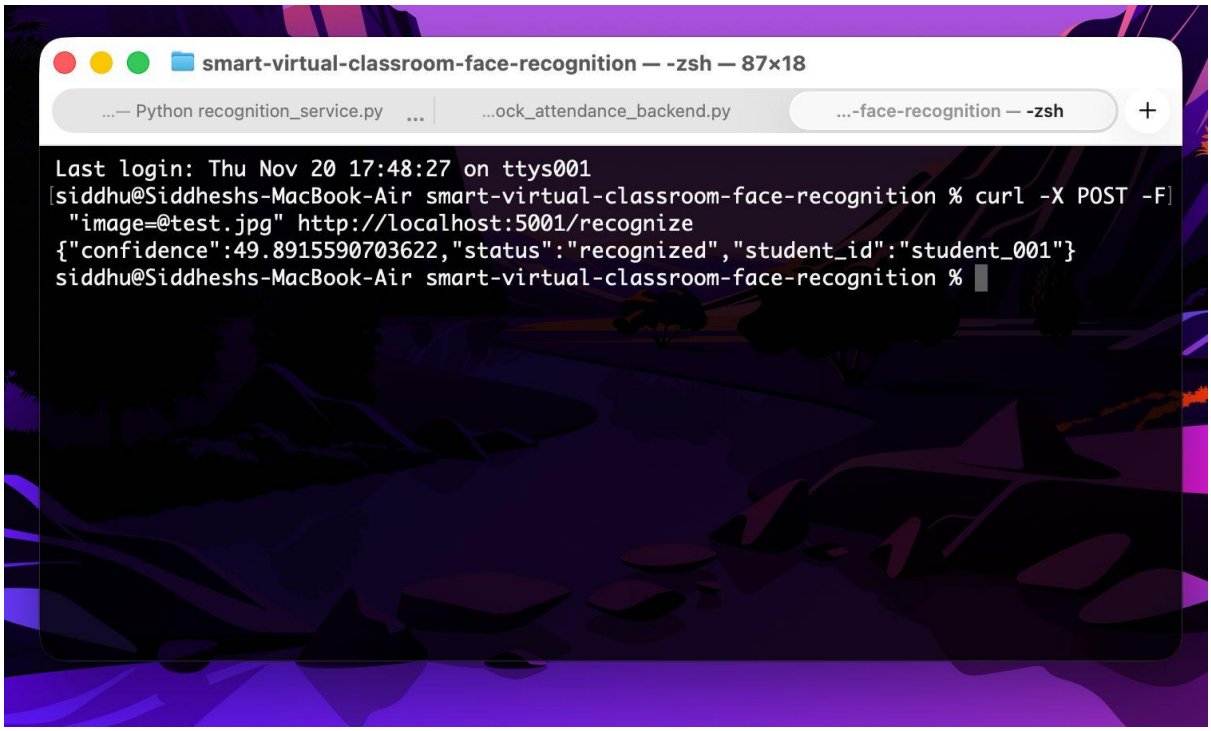
Model



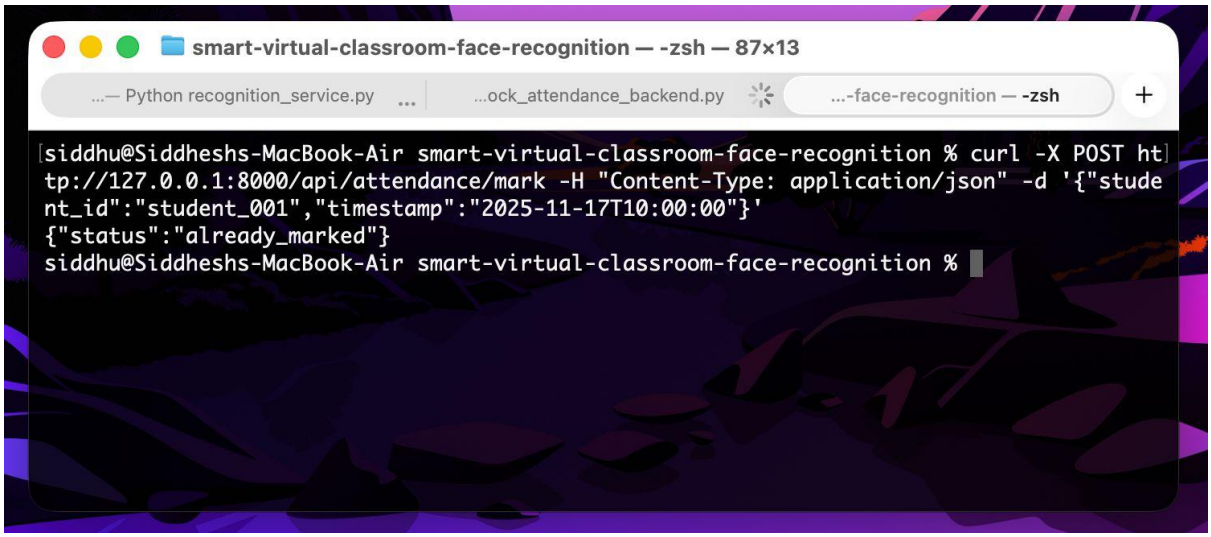
Project folder with test image



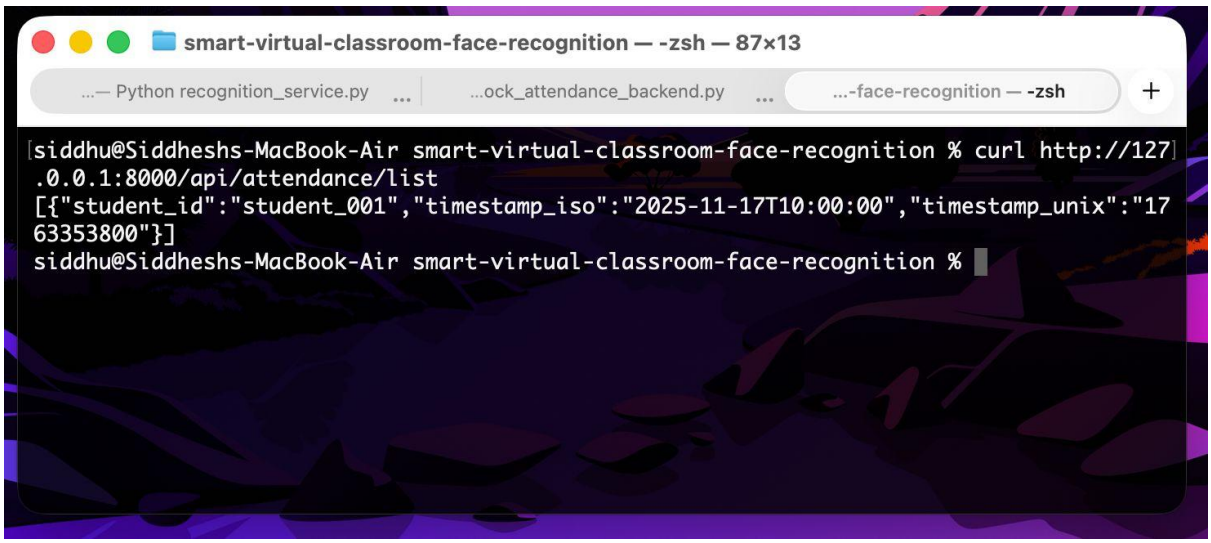
Testing the test image with curl -Post



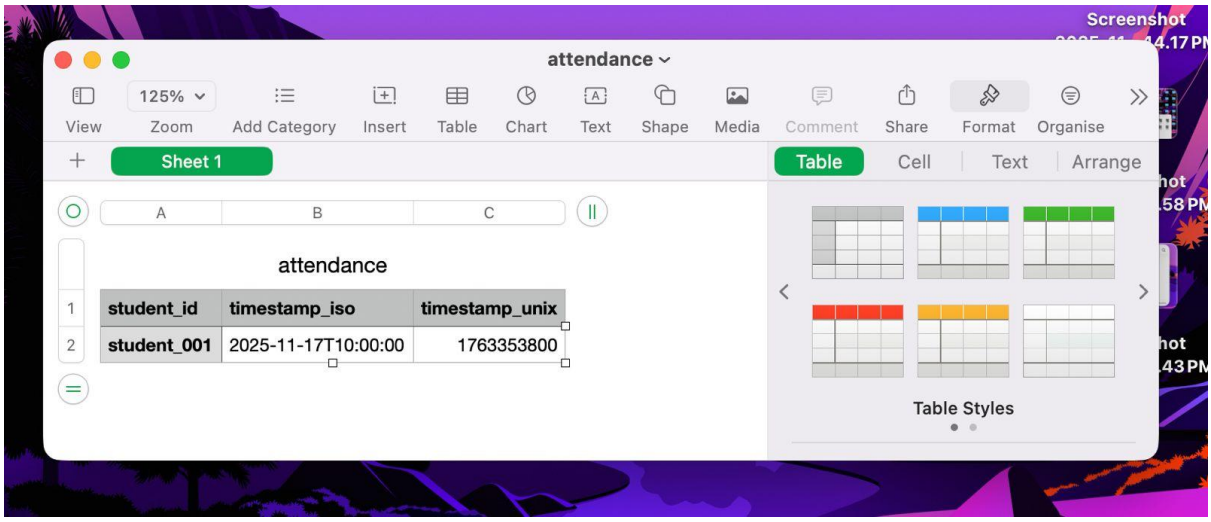
Marking attendance with Json



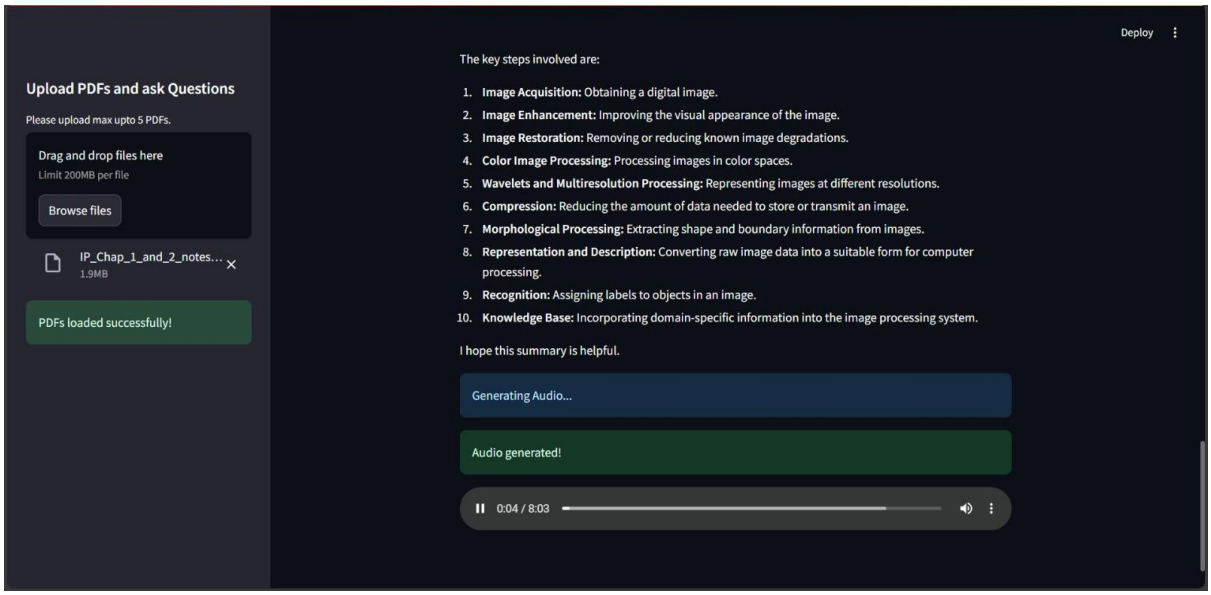
List of attendant students with help of curl

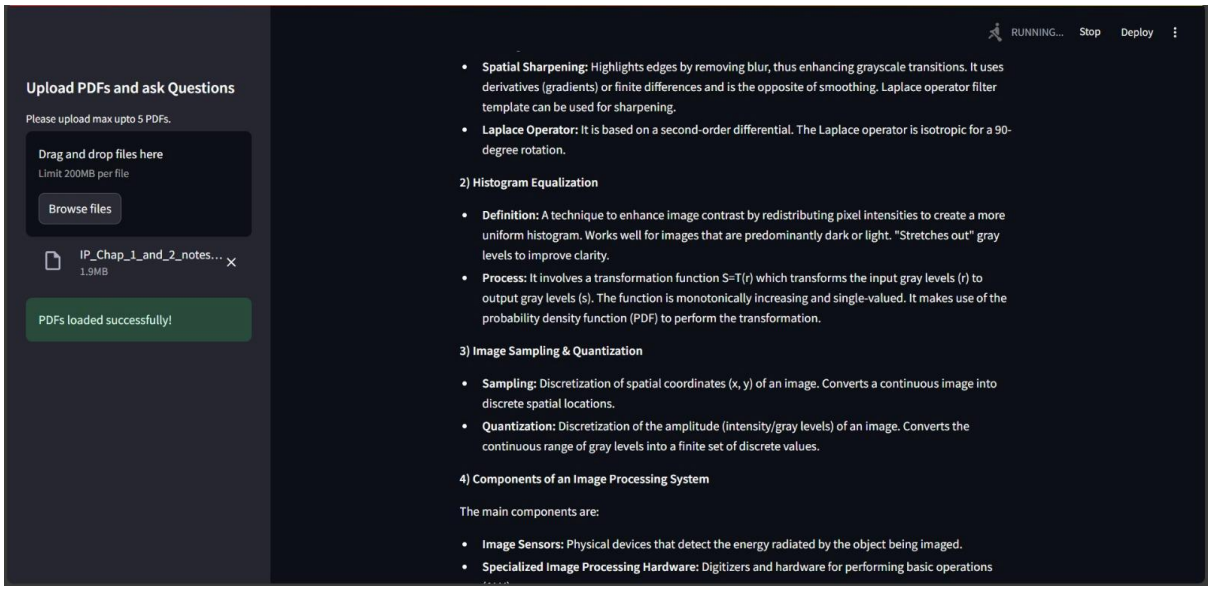


Csv file for database



2.PDF Summarizer





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IP_Chap_1_and_2_notes...
1.9MB

PDFs loaded successfully!

Response

Okay, here's a summary of the provided text, addressing each question in turn:

1) Spatial Smoothing & Sharpening

- **Spatial Domain Enhancement:** Image enhancement techniques that operate on the pixels within a defined neighborhood. The transformed intensity of a pixel is determined by the gray values of its neighbors. It's also called neighborhood operation or neighborhood processing.
- **Spatial Filtering:** Filtering in the spatial domain involves moving a filter (kernel, window) across the image. The filter's response at each point is calculated based on its contents and a predefined relationship ("template"). Linear spatial domain filtering uses linear operations in the neighborhood; nonlinear uses nonlinear operations.
- **Spatial Smoothing:** A technique to reduce noise in images. This is often achieved by averaging the pixel values in a neighborhood (average smoothing). Common smoothing filters include average smoothing, Gaussian smoothing, and adaptive smoothing.
  - **Average Smoothing:** Replaces each pixel's value with the average of its surrounding pixels. All pixels in the neighborhood contribute equally. It has the benefit of being simple but treats all the pixels in the neighborhood equally.
  - **Gaussian Smoothing:** Weights the center pixel of the neighborhood more heavily than distant pixels, giving a more natural smoothing effect. Reduces blur.
  - **Adaptive Smoothing:** Selectively smooths only in areas with noise, minimizing blurring. It uses local variance or the difference between maximum and minimum intensities to detect noisy regions.
- **Spatial Sharpening:** Highlights edges by removing blur, thus enhancing grayscale transitions. It uses derivatives (gradients) or finite differences and is the opposite of smoothing. Laplace operator filter

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PDFs loaded successfully!

4) Components of an Image Processing System

The main components are:

- **Image Sensors:** Physical devices that detect the energy radiated by the object being imaged.
- **Specialized Image Processing Hardware:** Digitizers and hardware for performing basic operations (ALU).
- **Computer:** A general-purpose computer to run the image processing software.
- **Software:** Specialized modules for performing specific image processing tasks.
- **Mass Storage:** System for storing large volumes of image data (short-term, online, archival).
- **Image Display:** Monitors for viewing images.
- **Hardcopy Devices:** Printers, cameras, etc., for producing hard copies of images.
- **Networking:** For transmitting image data between systems.

5) Basic Intensity Transformation Functions

The basic intensity transformation functions are:

- **Linear Transformations**
  - **Identity Transformation:** Output intensity is the same as the input intensity.
  - **Negative Transformation:** Inverts the intensity values, turning dark areas light and vice versa (S = L - 1 - r).
- **Logarithmic Transformations:**
  - **Log Transformation:** Compresses high intensity values and expands low intensity values. S = c log(r + 1). Enhances details in darker regions.
- **Power-Law (Gamma) Transformations:** S = c rgamma. Varies the intensity mapping based on the gamma value. Used for gamma correction in display devices.